6CloudCom Final Project

Members

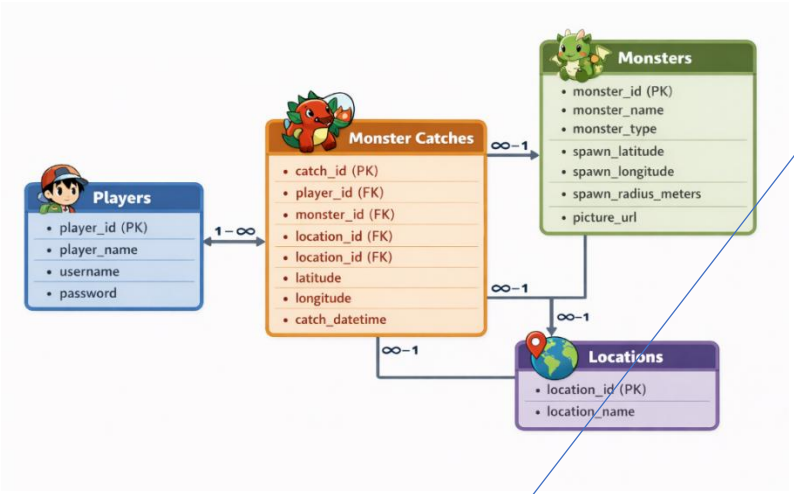
Date

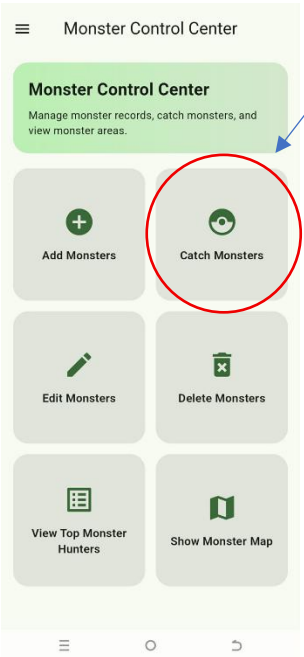
Final Project Requirement

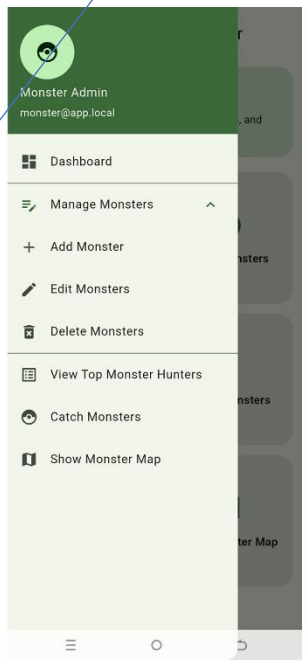
Design and build your cloud infrastructure that will support the following mobile application

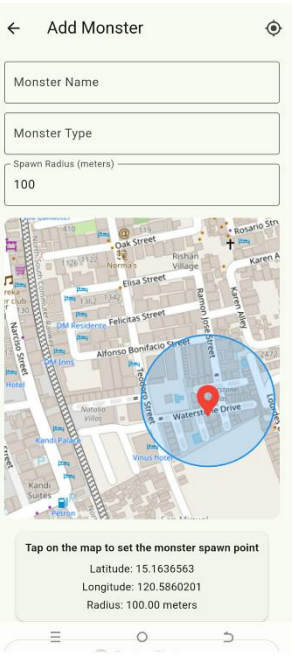
HAUPokemon Monster's App

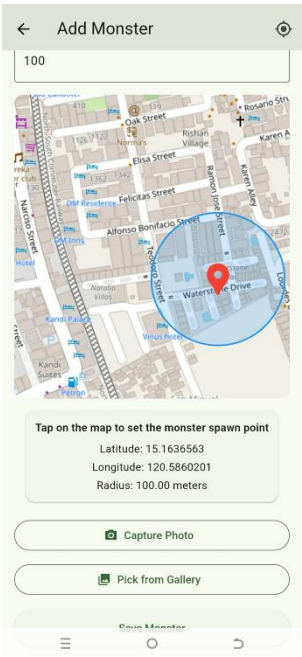
Include CRUD for Players, CRUD for Monsters, Top 10 Monster Hunter LeaderBoard and Catch Monsters

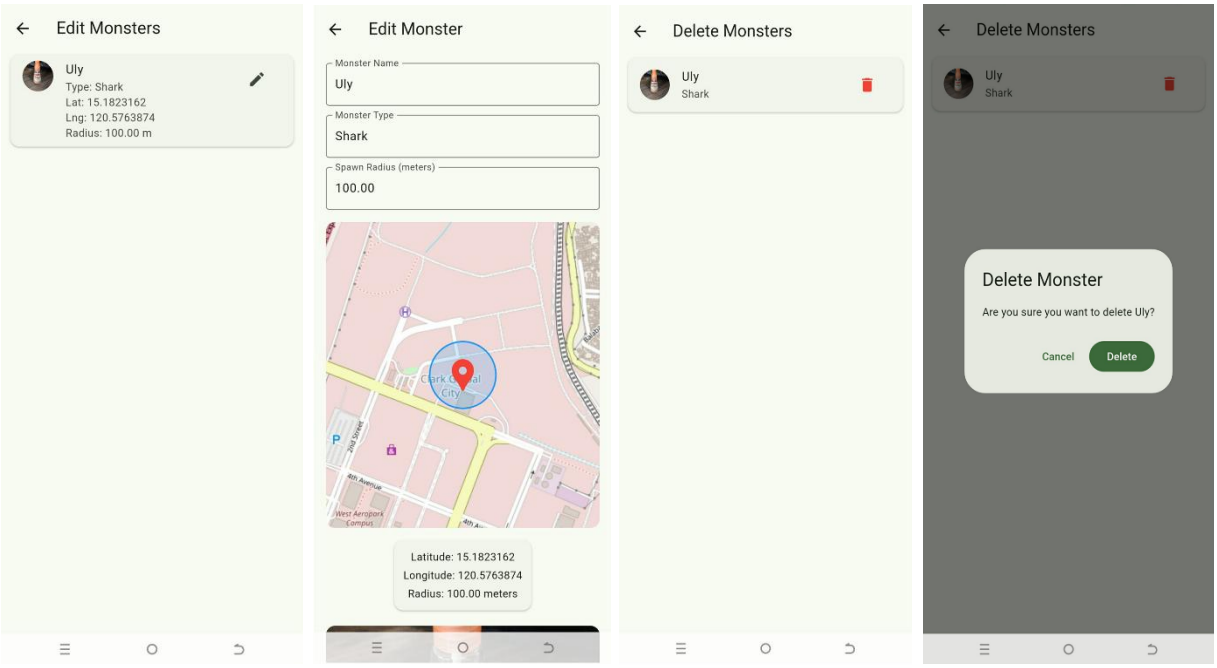












USE VPC Peering (Web Server to Database Server) and Tail Scale VPN when mobile phone connects to your Server.

Important: Only those connected thru VPN should be able to connect and transact with the web server

VPC Zones: North Virginia (Database)
Paris (Web Server)

Implement EC2 Automation Management ON/OFF on your mobile application.
and monitoring of EC2 instance if ON or OFF

DATABASE: haumonstersDB
TABLES:

Tables_in_haumonstersDB
locationstbl
monster_catchestbl
monsterstbl
playerstbl

monsterstbl:

Field	Type	Null	Key	Default	Extra
monster_id	int(10) unsigned	NO	PRI	NULL	auto_increment
monster_name	varchar(100)	NO		NULL	
monster_type	varchar(100)	NO		NULL	
spawn_latitude	decimal(10,7)	NO	MUL	NULL	
spawn_longitude	decimal(10,7)	NO		NULL	
spawn_radius_meters	decimal(10,2)	NO		100.00	
picture_url	varchar(500)	YES		NULL	

locationstbl:

Field	Type	Null	Key	Default	Extra
location_id	int(10) unsigned	NO	PRI	NULL	auto_increment
location_name	varchar(100)	NO		NULL	

monster_catchestbl:

Field	Type	Null	Key	Default	Extra
catch_id	int(10) unsigned	NO	PRI	NULL	auto_increment
player_id	int(10) unsigned	NO	MUL	NULL	
monster_id	int(10) unsigned	NO	MUL	NULL	
location_id	int(10) unsigned	NO	MUL	NULL	
latitude	decimal(10,7)	NO		NULL	
longitude	decimal(10,7)	NO		NULL	
catch_datetime	datetime	YES		current_timestamp()	

playerstbl:

Field	Type	Null	Key	Default	Extra
player_id	int(10) unsigned	NO	PRI	NULL	auto_increment
player_name	varchar(150)	NO		NULL	
username	varchar(100)	NO	UNI	NULL	
password	varchar(255)	NO		NULL	
created_at	datetime	YES		current_timestamp()	

HOW TO CATCH THE MONSTER:

During monster registration, you specify the spawn location (latitude and longtitude) using GPS.

When you click the button “Catch Monters” it will scan using the GPS if there is a registered monster within your location with the radius specified.

The Phone will sound an alarm if the monster is detected and will flash the torch light of the mobile phone for 5 seconds.

PLAYERS CRUD:

Create the interface for CRUD of players, they should have accounts and registration.

Field	Type	Null	Key	Default	Extra
player_id	int(10) unsigned	NO	PRI	NULL	auto_increment
player_name	varchar(150)	NO		NULL	
username	varchar(100)	NO	UNI	NULL	
password	varchar(255)	NO		NULL	
created_at	datetime	YES		current_timestamp()	

LEADER BOARD:

You will need to output the top 10 players with the most captured monsters.

SUBMISSION REQUIREMENTS:

1. APK / Web App
2. IAM CREDENTIALS FOR VIEWING
3. VIDEO RECORDING OF WORKING FEATURES (HIGHLIGHT DATABASE, MOBILE APP, VPN)
4. PRICING AND ESTIMATE (USE PRICING CALCULATOR)
5. NETWORK DIAGRAM

***Please see rubric**

PLEASE SEND A DIRECT MESSAGE THRU MS TEAMS IF YOU HAVE ANY INQUIRIES

RUBRIC:

Criteria	10	7	5	0
Overall Task Completion	Completed all task expected without misconfiguration, errors or bugs	Completed task expected with some misconfiguration, errors or bugs	Completed task with misconfiguration, or bugs	Not able to complete task expected
UI / UX Design (Includes About Us Page)	Supplied all the needed UI /UX content	20% of the expected design content are missing or incomplete	30% to 50% of the design UI/UX content are missing	Not able to supply the expected UI /UX design content
With Domain Name and Hosted on AWS EC2	The domain name is reachable regardless of device and the application is hosted on AWS	The domain name is reachable and the application is hosted on AWS but not responsive across devices	There is no domain name but the application is hosted on AWS	There is no domain name and the application is not hosted on AWS
Use of IAM, Security and Profiling	The application utilized IAM and has login module and with management of users	The application has login module but and management of users but not secure on the backend	The application has login module but does not have management of users	No login and no management of users
APK / Website Availability	The APK or Website App is working and available	The APK or Website is not complete	The APK or Website not working	No APK or Website

Implemented Serverless Computing with monitoring	This feature is avaible and working	This feature is available but incomplete	This feature is available but not working	Not available at all
VPN Connectivity (For Administrators)	The application is working perfectly with VPN especially for the backend	The application have VPN but have limitations	There is VPN setup but not working	Does not have VPN
Load Balancing and High-Availability	The application have load balancer and and a back-up mechanism for infrastructure and database	The application have load balancer but does not have a back-up mechanism for infrastructure and database	The application have load balancer and have a back-up mechanism for infrastructure and database but have limitations	There is no load balancer and a back-up mechanism for infrastructure and database configured.
Documentation (Pricing and Estimate, Demo Video of Application, Task List)	The expected output have complete documentation including the pricing and estimate of the project (e.g. domain registration, AWS infrastructure) and the video is consistent with the output and with task list	The expected output includes pricing and estimate (e.g. domain registration, AWS infrastructure) with task list, but not consistent with the video recording	The expected output includes pricing and estimate (e.g. domain registration, AWS infrastructure) with task list, but does not have video recording	No documentation at all
Submission Time and Checking	The group submitted earlier than expected due date.	The group submitted just on time	The group submitted late a day after of the original schedule	There is no submission